

Practice Questions for 2024-09-12-Duality

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1 Questions

1. Consider the primal problem (P): $\max \quad \xi(x) = c^T x$ subject to $Ax \leq b, x \geq 0$, and its dual (D): $\min \quad \xi(y) = b^T y$ subject to $A^T y \geq c, y \geq 0$. If the primal problem is feasible but unbounded, what can be concluded about the feasibility of the dual problem, and why?
 - a. The dual problem is also feasible and unbounded.
 - b. The dual problem is feasible and has an optimal solution.
 - c. The dual problem is infeasible.
 - d. The dual problem is unbounded.
 - e. There is not enough information to determine the feasibility of the dual problem.
2. Explain the concept of complementary slackness in the context of primal and dual linear programs. How does this concept relate to the optimality of solutions in both problems?
 - a. Complementary slackness states that the optimal objective function values of the primal and dual problems are equal.
 - b. Complementary slackness states that if a primal variable is positive, then the corresponding dual slack variable must be zero, and vice versa. Similarly, if a dual variable is positive, then the corresponding primal slack variable must be zero, and vice versa.
 - c. Complementary slackness states that if the primal problem is feasible, then the dual problem must also be feasible.
 - d. Complementary slackness states that the primal and dual problems always have optimal solutions.
 - e. Complementary slackness is only relevant for infeasible linear programs.
3. Given a primal linear program (P) and its dual (D), if both (P) and (D) are feasible, what can you conclude about the existence and values of their optimal solutions?
 - a. Both (P) and (D) must have optimal solutions, and their optimal objective function values are equal.
 - b. Both (P) and (D) must have optimal solutions, but their optimal objective function values may differ.
 - c. At least one of (P) or (D) must have an optimal solution, and their optimal objective function values are equal if both have optimal solutions.
 - d. Neither (P) nor (D) is guaranteed to have an optimal solution.

- e. If (P) has an optimal solution, then (D) must be infeasible.
4. Suppose you have a primal linear program (P) and its dual (D). You solve (P) using the simplex method and find that it is unbounded. What can you definitively conclude about the dual problem (D)? Explain your reasoning using the concept of weak duality.
- The dual problem (D) is also unbounded.
 - The dual problem (D) is infeasible.
 - The dual problem (D) has an optimal solution.
 - The dual problem (D) may be feasible or infeasible; there is not enough information to determine its status.
 - The dual problem (D) is unbounded above.
5. a. Consider the primal problem (P): $\max \xi(x) = c^T x$ subject to $Ax \leq b, x \geq 0$, and its dual (D): $\min \xi(y) = b^T y$ subject to $A^T y \geq c, y \geq 0$. If the primal problem (P) is feasible and has an optimal solution, what can we definitively conclude about the dual problem (D)?
- The dual problem (D) is infeasible.
 - The dual problem (D) is unbounded.
 - The dual problem (D) is feasible and has an optimal solution, with the same optimal objective function value as (P).
 - The dual problem (D) is feasible but may or may not have an optimal solution.
 - There is no relationship between the feasibility or optimality of (P) and (D).
- b. Explain the concept of weak duality and its implications for the relationship between the objective function values of feasible solutions to the primal and dual problems.
- Weak duality states that the objective function value of any feasible solution to the primal problem is greater than or equal to the objective function value of any feasible solution to the dual problem.
 - Weak duality states that the objective function value of any feasible solution to the primal problem is equal to the objective function value of any feasible solution to the dual problem.
 - Weak duality states that if the primal problem is feasible, then the dual problem must also be feasible.
 - Weak duality states that the objective function value of any feasible solution to the primal problem is less than or equal to the objective function value of any feasible solution to the dual problem.
 - Weak duality has no implications for the objective function values of feasible solutions to the primal and dual problems.
- c. Suppose the primal problem (P) is unbounded. What can we conclude about the dual problem (D) based on the duality theorems?
- The dual problem (D) is also unbounded.
 - The dual problem (D) is feasible and has an optimal solution.
 - The dual problem (D) is infeasible.
 - The dual problem (D) may be feasible or infeasible, but it cannot be unbounded.

- E. There is no relationship between the unboundedness of (P) and the properties of (D).
6. Consider the primal problem (P): $\max \xi(x) = c^T x$ subject to $Ax \leq b, x \geq 0$, and its dual (D): $\min \xi(y) = b^T y$ subject to $A^T y \geq c, y \geq 0$. If the primal problem is feasible but unbounded, what can be concluded about the feasibility of the dual problem, and why?
- The dual problem (D) is also feasible and unbounded.
 - The dual problem (D) is feasible and has an optimal solution.
 - The dual problem (D) is infeasible.
 - The dual problem (D) is unbounded.
 - There is not enough information to determine the feasibility of the dual problem.
7. Explain the concept of complementary slackness in the context of primal and dual linear programs. How does this concept relate to the optimality of solutions in both problems?
- Complementary slackness states that if a primal variable is positive, then the corresponding dual slack variable must be zero, and vice versa. This condition is necessary but not sufficient for optimality.
 - Complementary slackness states that the sum of the primal and dual objective functions must be zero at optimality. This condition is sufficient for optimality.
 - Complementary slackness states that the primal and dual problems must have the same optimal objective function value. This condition is necessary and sufficient for optimality.
 - Complementary slackness is a concept that only applies to unbounded linear programs. It states that the primal and dual problems are both unbounded.
 - Complementary slackness is irrelevant to the optimality of solutions in linear programming.
8. Given a primal linear program (P) and its dual (D), if both (P) and (D) are feasible, what can you conclude about the existence and values of their optimal solutions?
- Both (P) and (D) have optimal solutions, and their optimal objective function values are equal.
 - Both (P) and (D) have optimal solutions, but their optimal objective function values may differ.
 - At least one of (P) or (D) must be unbounded.
 - At least one of (P) or (D) must be infeasible.
 - There is not enough information to determine the existence or values of their optimal solutions.
9. Suppose you have a primal linear program (P) and its dual (D). You solve (P) using the simplex method and find that it is unbounded. What can you definitively conclude about the dual problem (D)? Explain your reasoning using the concept of weak duality.
- The dual problem (D) is also unbounded.
 - The dual problem (D) is feasible and has an optimal solution.
 - The dual problem (D) is infeasible.
 - The dual problem (D) is either infeasible or unbounded.

- e. There is not enough information to determine the feasibility or boundedness of the dual problem.
10. a. Consider the primal problem (P): $\max \quad \xi(x) = c^T x$ subject to $Ax \leq b, x \geq 0$, and its dual (D): $\min \quad \xi(y) = b^T y$ subject to $A^T y \geq c, y \geq 0$. If the primal problem is feasible but unbounded, what can be concluded about the feasibility of the dual problem, and why?
- A. The dual problem (D) is also feasible and unbounded.
 - B. The dual problem (D) is feasible and has an optimal solution.
 - C. The dual problem (D) is infeasible.
 - D. The dual problem (D) is unbounded.
 - E. There is not enough information to determine the feasibility of the dual problem.
- b. Explain the concept of complementary slackness in the context of primal and dual linear programs. How does this concept relate to the optimality of solutions in both problems?
- A. Complementary slackness states that the primal and dual objective functions are equal at optimality.
 - B. Complementary slackness states that if a primal constraint is binding, then the corresponding dual variable is zero, and vice versa.
 - C. Complementary slackness states that at optimality, the product of each primal variable and its corresponding dual slack variable is zero, and the product of each primal slack variable and its corresponding dual variable is zero.
 - D. Complementary slackness is only relevant for infeasible linear programs.
 - E. Complementary slackness states that the primal and dual problems must have the same number of variables.
- c. Given a primal linear program (P) and its dual (D), if both (P) and (D) are feasible, what can you conclude about the existence and values of their optimal solutions?
- A. Both (P) and (D) are unbounded.
 - B. Both (P) and (D) are infeasible.
 - C. Both (P) and (D) have optimal solutions, and their optimal objective function values are equal.
 - D. Either (P) or (D) has an optimal solution, but not both.
 - E. There is not enough information to determine the existence of optimal solutions.

2 Answers

1. Consider the primal problem (P): $\max \quad \xi(x) = c^T x$ subject to $Ax \leq b, x \geq 0$, and its dual (D): $\min \quad \xi(y) = b^T y$ subject to $A^T y \geq c, y \geq 0$. If the primal problem is feasible but unbounded, what can be concluded about the feasibility of the dual problem, and why?
 - a. If the primal is unbounded, the dual cannot be feasible, let alone unbounded. Weak duality would be violated.
 - b. Similar to A, if the primal is unbounded, the dual cannot be feasible.
 - c. The dual problem is infeasible. If the primal problem is feasible and unbounded, it means that the objective function can increase without bound while satisfying the constraints. By weak duality, the objective function value of any feasible solution to the primal problem is less than or equal to the objective function value of any feasible solution to the dual problem. If the primal objective function is unbounded, then the dual problem cannot have a feasible solution, otherwise it would contradict weak duality.
 - d. The dual problem cannot be unbounded if the primal is unbounded. Weak duality would be violated.
 - e. Weak duality provides sufficient information to determine the feasibility of the dual problem in this case.
2. Explain the concept of complementary slackness in the context of primal and dual linear programs. How does this concept relate to the optimality of solutions in both problems?
 - a. This describes strong duality, not complementary slackness.
 - b. Complementary slackness states that for each pair of primal and dual variables, at least one must be zero at optimality. Specifically, if a primal variable x_j is positive ($x_j > 0$), then the corresponding dual slack variable z_j must be zero ($z_j = 0$), and vice versa. Similarly, if a dual variable y_i is positive ($y_i > 0$), then the corresponding primal slack variable w_i must be zero ($w_i = 0$), and vice versa. This condition is necessary and sufficient for optimality in both the primal and dual problems.
 - c. This is a statement about feasibility, not complementary slackness.
 - d. This is false; primal and dual problems can be infeasible or unbounded.
 - e. Complementary slackness is a condition for optimality, applicable to feasible problems.
3. Given a primal linear program (P) and its dual (D), if both (P) and (D) are feasible, what can you conclude about the existence and values of their optimal solutions?
 - a. If both the primal and dual problems are feasible, then both must have optimal solutions, and their optimal objective function values are equal. This is a direct consequence of the strong duality theorem.
 - b. The optimal objective function values must be equal due to strong duality.
 - c. Both problems are guaranteed to have optimal solutions if both are feasible.
 - d. This is incorrect; the strong duality theorem guarantees the existence of optimal solutions under the given condition.
 - e. This is incorrect; if the primal has an optimal solution, the dual also has an optimal solution with the same objective value.

4. Suppose you have a primal linear program (P) and its dual (D). You solve (P) using the simplex method and find that it is unbounded. What can you definitively conclude about the dual problem (D)? Explain your reasoning using the concept of weak duality.
- If the primal is unbounded, the dual must be infeasible. An unbounded primal implies no feasible dual.
 - The dual problem (D) is infeasible. Weak duality states that for any feasible solution x to (P) and any feasible solution y to (D), the objective function value of (P) is less than or equal to the objective function value of (D): $c^T x \leq b^T y$. If (P) is unbounded, meaning its objective function can be made arbitrarily large, then there cannot exist any feasible solution y to (D). Otherwise, we would have a contradiction: an arbitrarily large value on the left side of the inequality, but a finite value on the right side.
 - If the primal is unbounded, the dual cannot have a solution.
 - Weak duality allows us to conclude the dual is infeasible.
 - The dual is a minimization problem, so it cannot be unbounded above. It would be unbounded below if it were unbounded.
5. a. Consider the primal problem (P): $\max \xi(x) = c^T x$ subject to $Ax \leq b, x \geq 0$, and its dual (D): $\min \xi(y) = b^T y$ subject to $A^T y \geq c, y \geq 0$. If the primal problem (P) is feasible and has an optimal solution, what can we definitively conclude about the dual problem (D)?
- This is incorrect. The Strong Duality Theorem establishes a direct relationship between the optimality of the primal and dual problems. If the primal has an optimal solution, the dual must also have one.
 - This is incorrect. If the primal problem is feasible and has an optimal solution, it cannot be unbounded. The dual problem would also have an optimal solution, not be unbounded.
 - The Strong Duality Theorem states that if the primal problem (P) has an optimal solution, then the dual problem (D) also has an optimal solution, and their optimal objective function values are equal. Since (P) is feasible and has an optimal solution, (D) must also be feasible and have an optimal solution with the same optimal objective value.
 - This is incorrect. The Strong Duality Theorem guarantees that if the primal has an optimal solution, the dual also has an optimal solution.
 - This is incorrect. The Strong Duality Theorem explicitly links the optimality of the primal and dual problems.
- b. Explain the concept of weak duality and its implications for the relationship between the objective function values of feasible solutions to the primal and dual problems.
- This is incorrect. This statement reverses the inequality in the definition of weak duality.
 - This is incorrect. This statement describes strong duality, not weak duality. Weak duality only provides an inequality, not an equality.
 - This is incorrect. Weak duality does not guarantee the feasibility of the dual problem if the primal is feasible. It only relates the objective function values of feasible solutions.
 - Weak duality states that for any feasible solution to the primal problem and any feasible solution to the dual problem, the objective function value of the primal solution is less than or equal to the objective function value of the dual solution. This implies that the optimal objective function value of the primal problem is always less than or equal to the optimal objective function value of the dual problem.
 - This is incorrect. Weak duality provides a fundamental relationship between the objective function values of feasible solutions to the primal and dual problems.

- c. Suppose the primal problem (P) is unbounded. What can we conclude about the dual problem (D) based on the duality theorems?
- A. This is incorrect. If the primal is unbounded, the dual must be infeasible.
 - B. This is incorrect. An unbounded primal implies an infeasible dual.
 - C. If the primal problem (P) is unbounded, then the dual problem (D) must be infeasible. This is a direct consequence of weak duality, which states that the objective function value of any feasible solution to the primal problem is less than or equal to the objective function value of any feasible solution to the dual problem. If (P) is unbounded, its objective function can be made arbitrarily large, which would contradict weak duality if (D) were feasible.
 - D. This is incorrect. The dual problem must be infeasible.
 - E. This is incorrect. There is a direct relationship between the unboundedness of the primal and the infeasibility of the dual, as established by duality theorems.
6. Consider the primal problem (P): $\max \quad \xi(x) = c^T x$ subject to $Ax \leq b, x \geq 0$, and its dual (D): $\min \quad \xi(y) = b^T y$ subject to $A^T y \geq c, y \geq 0$. If the primal problem is feasible but unbounded, what can be concluded about the feasibility of the dual problem, and why?
- a. Incorrect. Weak duality states that the primal objective function is always less than or equal to the dual objective function for feasible solutions. If the primal is unbounded, the dual cannot be feasible.
 - b. Incorrect. If the primal is unbounded, the dual cannot have a finite optimal solution.
 - c. The correct answer is C. If the primal problem (P) is feasible but unbounded, then the dual problem (D) must be infeasible. This is a direct consequence of weak duality, which states that for any feasible solutions x and y to the primal and dual problems, respectively, $c^T x \leq b^T y$. If the primal objective function $c^T x$ can be made arbitrarily large (unbounded), then the dual objective function $b^T y$ must also be unbounded above, which is a contradiction since the dual is a minimization problem. Therefore, the dual problem must be infeasible.
 - d. Incorrect. An unbounded dual would imply an infeasible primal.
 - e. Incorrect. The feasibility of the dual problem can be definitively determined from the unboundedness of the primal problem using weak duality.
7. Explain the concept of complementary slackness in the context of primal and dual linear programs. How does this concept relate to the optimality of solutions in both problems?
- a. Complementary slackness states that for an optimal solution, if a primal variable x_j is strictly positive ($x_j > 0$), then the corresponding dual slack variable z_j must be zero ($z_j = 0$), and vice versa. Similarly, if a dual variable y_i is strictly positive ($y_i > 0$), then the corresponding primal slack variable w_i must be zero ($w_i = 0$), and vice versa. This condition, along with primal and dual feasibility, is necessary and sufficient for optimality. It essentially states that at optimality, resources are either fully utilized (primal variable positive, slack zero) or have excess capacity (primal variable zero, slack positive).
 - b. Incorrect. The sum of the primal and dual objective functions is not directly related to complementary slackness. Complementary slackness deals with the relationship between individual primal and dual variables and their corresponding slack variables.
 - c. Incorrect. While the primal and dual problems have the same optimal objective function value at optimality (strong duality), this is a separate concept from complementary slackness. Complementary slackness deals with the relationship between individual variables and their slacks.

- d. **Incorrect.** Complementary slackness applies to both bounded and unbounded linear programs. In the case of unboundedness, complementary slackness is not satisfied.
 - e. **Incorrect.** Complementary slackness is a fundamental concept in linear programming and is crucial for determining the optimality of solutions.
8. Given a primal linear program (P) and its dual (D), if both (P) and (D) are feasible, what can you conclude about the existence and values of their optimal solutions?
- a. If both the primal (P) and dual (D) linear programs are feasible, then both problems have optimal solutions, and their optimal objective function values are equal. This is a direct consequence of the strong duality theorem. The strong duality theorem states that if a primal problem has an optimal solution, then the dual problem also has an optimal solution, and their optimal objective function values are equal. The converse is also true. Since both are feasible, the condition for the strong duality theorem is met.
 - b. **Incorrect.** The strong duality theorem states that if both primal and dual are feasible, their optimal objective function values are equal.
 - c. **Incorrect.** If both primal and dual are feasible, neither can be unbounded.
 - d. **Incorrect.** The premise states that both are feasible.
 - e. **Incorrect.** The strong duality theorem provides a definitive conclusion about the existence and equality of optimal objective function values.
9. Suppose you have a primal linear program (P) and its dual (D). You solve (P) using the simplex method and find that it is unbounded. What can you definitively conclude about the dual problem (D)? Explain your reasoning using the concept of weak duality.
- a. **Incorrect.** Weak duality implies that if the primal is unbounded, the dual must be infeasible.
 - b. **Incorrect.** If the primal is unbounded, the dual cannot have a finite optimal solution.
 - c. The correct answer is C. If the primal problem (P) is unbounded, then the dual problem (D) must be infeasible. Weak duality states that for any feasible solutions x and y to the primal and dual problems, respectively, $c^T x \leq b^T y$. If (P) is unbounded, then there exists a sequence of feasible solutions $\{x_k\}$ such that $c^T x_k \rightarrow \infty$ as $k \rightarrow \infty$. If (D) were feasible, then there would exist a feasible solution y such that $c^T x_k \leq b^T y$ for all k . However, this would imply that $b^T y$ is unbounded above, which is impossible for a minimization problem. Therefore, (D) must be infeasible.
 - d. **Incorrect.** The dual problem cannot be unbounded if the primal is unbounded.
 - e. **Incorrect.** Weak duality provides a definitive conclusion about the infeasibility of the dual problem.
10. a. Consider the primal problem (P): $\max \xi(x) = c^T x$ subject to $Ax \leq b, x \geq 0$, and its dual (D): $\min \xi(y) = b^T y$ subject to $A^T y \geq c, y \geq 0$. If the primal problem is feasible but unbounded, what can be concluded about the feasibility of the dual problem, and why?
- A. **Incorrect.** Weak duality states that the primal objective function is always less than or equal to the dual objective function for feasible solutions. An unbounded primal problem contradicts this if the dual is feasible.
 - B. **Incorrect.** If the primal is unbounded, the dual cannot have an optimal solution because the primal objective function can exceed any finite value of the dual objective function.

- C. The correct answer is C. If the primal problem (P) is feasible but unbounded, then the dual problem (D) must be infeasible. This is a direct consequence of weak duality, which states that for any feasible solutions x and y to the primal and dual problems, respectively, $c^T x \leq b^T y$. If the primal objective function can be made arbitrarily large (unbounded), then the dual problem cannot have any feasible solutions, otherwise the inequality would be violated.
- D. **Incorrect.** An unbounded dual problem would imply an infeasible primal problem, which is the opposite of the given condition.
- E. **Incorrect.** The feasibility of the dual problem is directly determined by the boundedness of the primal problem due to weak duality.
- b. Explain the concept of complementary slackness in the context of primal and dual linear programs. How does this concept relate to the optimality of solutions in both problems?
- A. **Incorrect.** While the primal and dual objective functions are equal at optimality (strong duality), this is not the definition of complementary slackness.
- B. **Incorrect.** This describes a partial aspect of complementary slackness but doesn't encompass the full condition involving primal slack variables and dual variables.
- C. The correct answer is C. Complementary slackness is a necessary and sufficient condition for optimality in linear programming. It states that for an optimal primal solution and an optimal dual solution, the product of each primal variable and its corresponding dual slack variable must be zero, and the product of each primal slack variable and its corresponding dual variable must be zero. This means that if a primal variable is positive, its corresponding dual slack variable must be zero, and vice versa. Similarly, if a primal slack variable is positive, its corresponding dual variable must be zero, and vice versa.
- D. **Incorrect.** Complementary slackness is a fundamental concept applicable to feasible linear programs with optimal solutions.
- E. **Incorrect.** Complementary slackness does not impose any restrictions on the number of variables in the primal and dual problems; it relates to the values of the variables and slack variables at optimality.
- c. Given a primal linear program (P) and its dual (D), if both (P) and (D) are feasible, what can you conclude about the existence and values of their optimal solutions?
- A. **Incorrect.** If both primal and dual are feasible, they cannot be unbounded. Unboundedness in one implies infeasibility in the other.
- B. **Incorrect.** The premise states that both are feasible, meaning they have at least one solution satisfying all constraints.
- C. The correct answer is C. If both the primal problem (P) and its dual (D) are feasible, then both problems have optimal solutions, and their optimal objective function values are equal. This is a direct consequence of the strong duality theorem. The strong duality theorem states that if a primal linear program has an optimal solution, then its dual also has an optimal solution, and the optimal objective function values are equal. The converse is also true.
- D. **Incorrect.** The strong duality theorem guarantees that if one has an optimal solution, the other does as well, and their optimal values are equal.
- E. **Incorrect.** The strong duality theorem provides a definitive conclusion about the existence and equality of optimal solutions when both primal and dual are feasible.